

Problem Set 0

CSE5825 BML

1 Probability

1. You run a petting zoo that has many goats. Some goats are mostly nice and very rarely bite, while other goats are mean and are more prone to biting. Assume we believe that $P(\text{Bite}|\text{mean}) = 0.8$ and we learn of an instance where a goat bit a child.
 - (a) What additional assumptions (if any) do we need to make to compute the probability that the biting goat is mean?
 - (b) Given those assumptions, what is the probability that the biting goat is mean?
2. You are given the graphical model in Figure 1. Consider another random variable W .

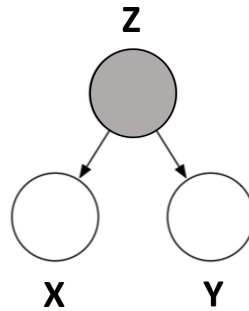


Figure 1: A simple graphical model.

- (a) Prove that $(X \perp\!\!\!\perp Y, W|Z) \Rightarrow (X \perp\!\!\!\perp Y|Z)$.
- (b) Prove that $(X \perp\!\!\!\perp Y, W|Z) \Rightarrow (X \perp\!\!\!\perp Y|W, Z)$

2 A simple model

Given a simple directed graphical model consisting of two nodes A and B and a single edge from A to B . The joint distribution is given in Figure 2. The *MAP* is the probability of the most likely assignment to all non-evidence variables.

1. What is $MAP(A)$ (i.e. $\arg \max_a(P(a))$)?
2. What is $MAP(B)$? (i.e. $\arg \max_b(P(b))$)?
3. What is $MAP(A, B)$? (i.e. $\arg \max_{a,b}(P(a, b))$)?

a^0	a^1		
0.4	0.6		

A	b^0	b^1
a^0	0.1	0.9
a^1	0.5	0.5

Figure 2: Joint distribution for a simple $A \rightarrow B$ graphical model.

3 Conditional independencies

Given the graphical model in Figure 3:

1. prove that $(S \perp\!\!\!\perp D, G, L | I)$.
2. is $(D \perp\!\!\!\perp I | G)$ true?
3. is $(D \perp\!\!\!\perp I)$ true?
4. is $(D \perp\!\!\!\perp I | L)$ true?

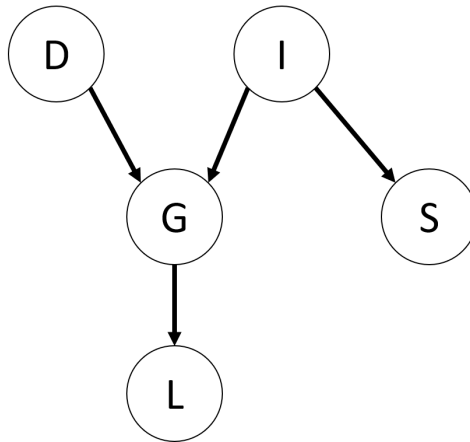


Figure 3: A DGM on 5 random variables.

4 A million dollar app idea

You are an aspiring cell phone app maker and you think you have an idea that will make you millions. Your app classifies text messages that you receive as either (a) spam, (b) from a relative that you avoid at family parties, or (c) anything else and allows the user to automatically delete messages that are classified as (a) or (b). You develop the model in Figure 4 for this problem. Despite the strong assumptions this model makes about conditional independence, you think this model might work.

Let C denote the *class* random variable and $X_1, \dots, X_n$ the observed random variables.

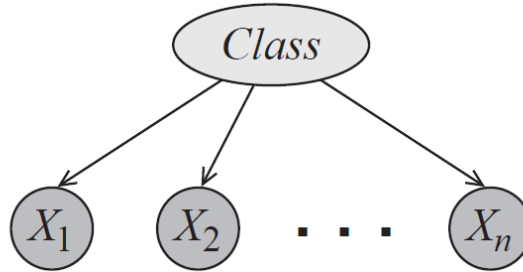


Figure 4: A naive graphical model.

1. How is C distributed?
2. How are the X_i for $i = 1, \dots, n$ distributed?
3. Factorize the joint probability $P(C, X_1, \dots, X_n)$.
4. What distribution does the posterior follow?
5. Is this a good model for your particular problem?

5 CpG Islands

You are given a sequence of DNA where observations $X \in \{A, C, G, T\}$. Regions of the genome that are involved in the regulation of genes are rich in G and C , and have probability of $(0.15, 0.35, 0.35, 0.15)$ for emitting characters (A, C, G, T) . Non-regulatory regions have probability of $(0.25, 0.25, 0.25, 0.25)$ for emitting characters (A, C, G, T) . Assume that you are equally likely to start in a regulatory or non-regulatory region and that you have probability of 0.9 to stay within a region and 0.1 to switch between regions.

1. Develop a Hidden Markov Model for this problem.
2. Implement the sum-product and max-product algorithms for inference in your HMM.
3. Run it on the section of human chromosome 1 DNA in HuskyCT.
4. How much of the DNA is identified as within a CpG island when considering (a) the maximum likelihood path through the HMM and (b) the maximum marginal probability at each DNA site?
5. Can you think of any other models (HMM or otherwise) that can model this problem?